CS6550 — Database Management System (advanced, with CS3700)

Tier	B
Phase (order)	6
Do this first	CS3700
Estimated effort	10-12 weeks
Unlocks	CS6510, CS5830

Roadmap

1. Query processing and optimisation, indexes (B+-tree, hash).
2. Transactions, concurrency control, recovery (ARIES).

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- CMU Advanced Database Systems (15-721 / Spring 2024) (22 videos). CMU 15-721 *Advanced Database Systems* (Spring 2024, Andy Pavlo): modern OLAP/column-store internals.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- CMU Intro to Database Systems (15-445/645 - Fall 2025) (25 videos). CMU 15-445 Fall 2025: the intro course if you skipped it.
- Distributed Systems lecture series (23 videos). Martin Kleppmann *Distributed Systems* (Cambridge): replication + consistency for distributed databases.

⚡ Short videos for intuition and revision

- Designing Data Intensive Applications (11 videos). *Designing Data-Intensive Applications* chapter walkthroughs (Ben Dicken).

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–4	Modern OLAP systems, columnar storage and encoding. Read the Parquet format docs.
2	5–9	Query execution, vectorisation with SIMD, JIT compilation, scheduling.

Week	Lectures	What to do
3	10–12	Parallel hash joins, worst-case optimal joins, UDF optimisation.
4	13–16	Networking protocols, query optimiser implementation.
5	17–22	System case studies: BigQuery, Photon, Snowflake, DuckDB, Yellowbrick, Redshift. Read one paper per system; benchmark DuckDB vs Postgres on TPC-H.

▼ **All lectures in *CMU Advanced Database Systems (15-721 / Spring 2024)* (22 videos)**

1. S2024 #00 - Course Overview & Logistics (CMU Advanced Database Systems)
2. S2024 #01 - Modern OLAP Database Systems (CMU Advanced Database Systems)
3. S2024 #02 - Data Formats & Encoding Part 1 (CMU Advanced Database Systems)
4. S2024 #03 - Data Formats & Encoding Part 2 (CMU Advanced Database Systems)
5. S2024 #04 - Query Execution & Processing Part 1 (CMU Advanced Database Systems)
6. S2024 #05 - Query Execution & Processing Part 2 (CMU Advanced Database Systems)
7. S2024 #06 - Vectorized Query Execution Using SIMD (CMU Advanced Database Systems)
8. S2024 #07 - JIT Query Compilation & Code Generation (CMU Advanced Database Systems)
9. S2024 #08 - Query Scheduling & Coordination (CMU Advanced Database Systems)
10. S2024 #09 - Parallel Hash Join Algorithms (CMU Advanced Database Systems)
11. S2024 #10 - Multi-Way Join Algorithms / Worst-Case Optimal Joins (CMU Advanced Database Systems)
12. S2024 #11 - User-Defined Function Optimizations (CMU Advanced Database Systems)
13. S2024 #12 - Database Networking Protocols (CMU Advanced Database Systems)
14. S2024 #13 - Query Optimizer Implementation 1 (CMU Advanced Database Systems)
15. S2024 #14 - Query Optimizer Implementation 2 (CMU Advanced Database Systems)
16. S2024 #15 - Query Optimizer Implementation 3 (CMU Advanced Database Systems)
17. S2024 #17 - Google BigQuery / Dremel (CMU Advanced Database Systems)
18. S2024 #18 - Databricks Photon / Spark SQL (CMU Advanced Database Systems)
19. S2024 #19 - Snowflake Data Warehouse Internals (CMU Advanced Database Systems)
20. S2024 #20 - DuckDB Embedded Database System (CMU Advanced Database Systems)
21. S2024 #21 - Yellowbrick Data Warehouse System (CMU Advanced Database Systems)
22. S2024 #22 - Amazon Redshift Data Warehouse System (CMU Advanced Database Systems)

Books, notes and websites

- CMU 15-445 / 15-721 lectures
- Book: *Designing Data-Intensive Applications* (Kleppmann)
- Book: Ramakrishnan & Gehrke

Practice / proof of understanding

Build a toy key-value store with a B+-tree and a write-ahead log.

What your IITM course sheet says

From the PDF: CS6550

Official description: Need for database approach; data independence; data models; hierarchical, network and relational; TMS architecture; network approach; DBTG report, scheme and subscheme, currency, data manipulation language; relational models; join and projection operations, relational algebra and calculus, normalization, query languages, database administrator, security and integrity considerations, compression techniques.

Official textbooks: NIL